Assessment of Low-Carbon Development Technologies in the Iron and Steel Industry

SAUDI ARAMCO & Tsinghua University

清华大学气候变化与可持续发展研究院
Institute of Climate Change and Sustainable Development,Tsinghua University

Presenter: Yuezhang He, PhD candidate

Research team: Prof. Zheng Li, Prof. Tianduo Peng, Yuezhang He, Zhenqian Wang

Oct 14, 2024

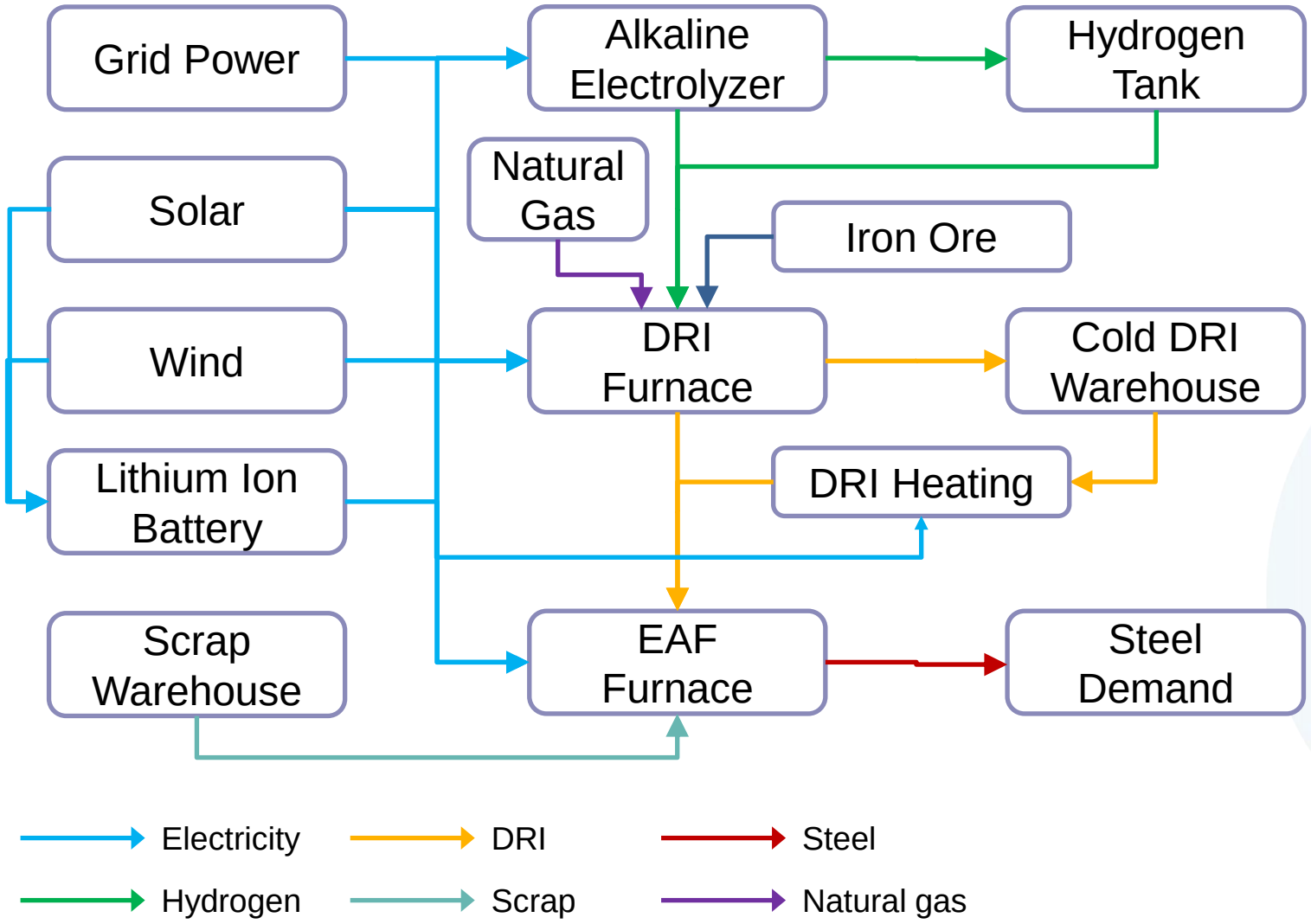
Project Timeline

	Nov	Dec	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr
Research Proposal & preparations																		
Overview of China's iron & steel industry																		
Data requirement & database investigation																		
Technology assessment																		
Accounting models & technology report									1									
Investigation on breakthrough technology industry chain																		
High spatial resolution data collation																		
Model development																		
Data analysis & Scenario design																		
Final report																		2

➢ Milestones

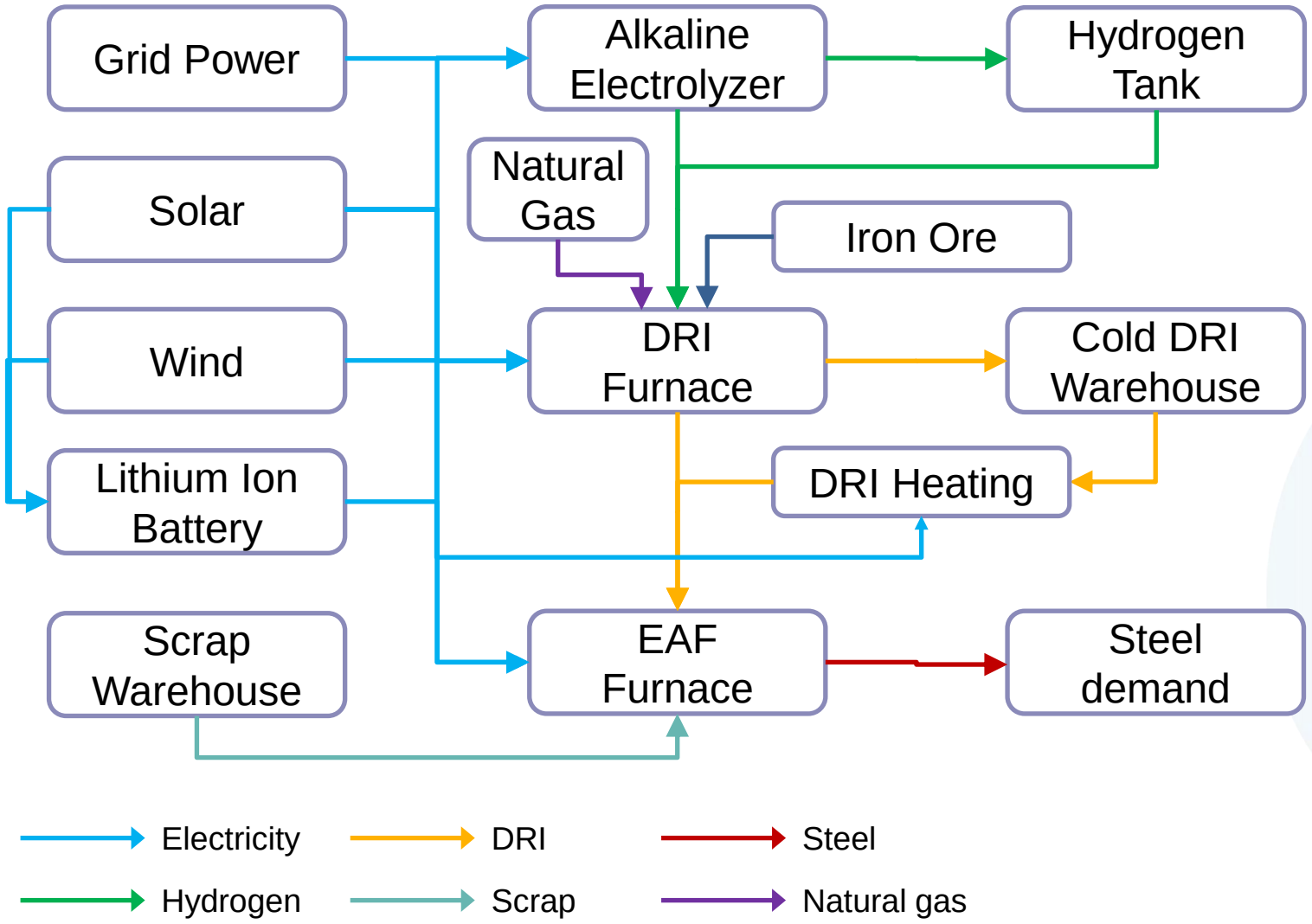
- 1. Assessment of Low-Carbon Development Technologies in the iron and steel industry: Model & Report
- 2. Study on the Evolution of the Industrial Layout of the iron and steel industry: Model & Report

A general EAF framework with flexibility



- Optimal configuration:
  - Solar/wind
  - Lithium-ion Battery
  - Electrolyzer
  - Storage tank
  - DRI furnace (flexible)
  - Iron ore
  - EAF furnace (flexible)
  - DRI warehouse
  - Scrap warehouse
  - Steel
  - Grid power
  - Natural gas blend

A general EAF framework with flexibility



	Flexibility Operation	Buffer
H_2	Electrolyzer	Battery
DRI	DRI furnace	Battery Storage tank
Crude Steel	EAF furnace (input)	Battery DRI warehouse scrap warehouse
Crude Steel	EAF furnace (output)	Steel demand
Reductant	DRI furnace	Natural gas
Power Source	/	Grid power

Scenario design

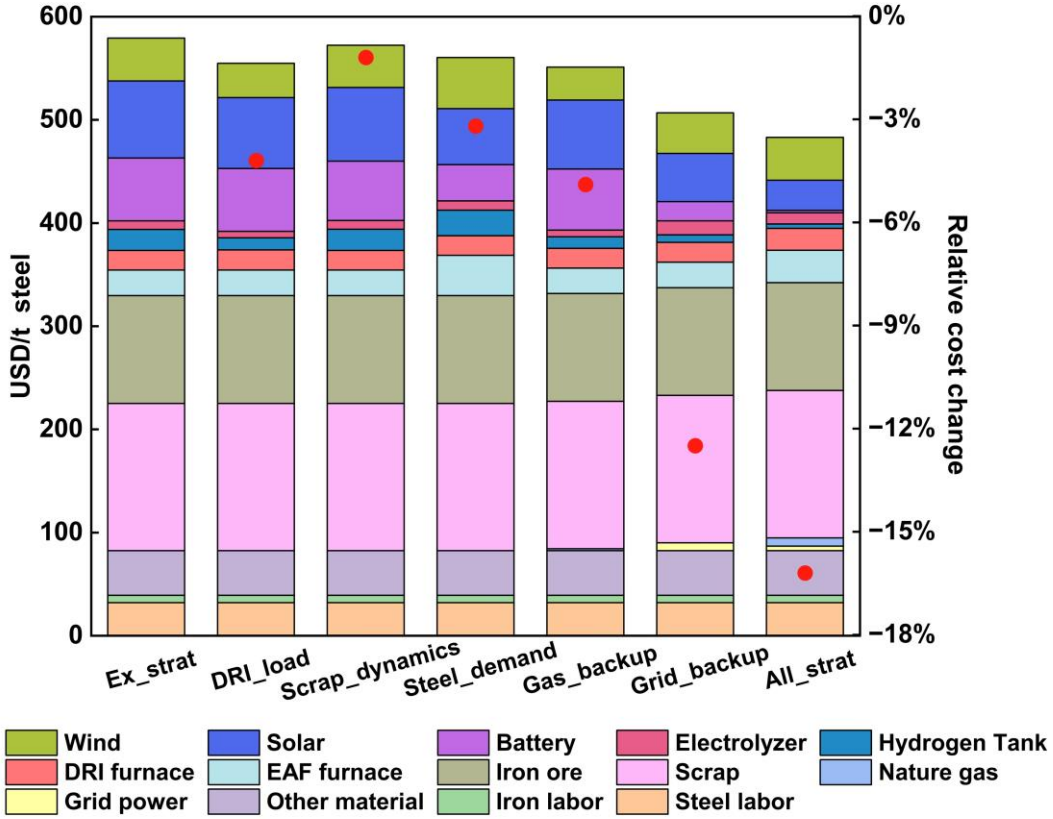
	Flexibility Operation						
Scenario	**Electrolyzer**	**EAF load**	**DRI load**	**Scrap blend ratio**	**Steel demand**	**Gas backup**	**Grid backup**
Ex_strat	Yes	Yes	No	No	No	No	No
DRI_load	Yes	Yes	Yes	No	No	No	No
Scrap_dynamics	Yes	Yes	No	Yes	No	No	No
Steel_demand	Yes	Yes	No	No	Yes	No	No
Gas_backup	Yes	Yes	No	No	No	Yes	No
Grid_backup	Yes	Yes	No	No	No	No	Yes
All_flexibility	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Assumptions:

1. The ratio of scrap is set to be 50%, and scrap cost is $286/t.
2. The renewable output curve adopts data in Hebei; The steel demand is assumed as 10 Mt/year.
3. Wind, solar, battery, and electrolyzer use projected data in 2035.
4. The emission cap is set as 0.1 tCO_2/t steel, which is ~5% of BF-BOF.

Crude steel cost structure

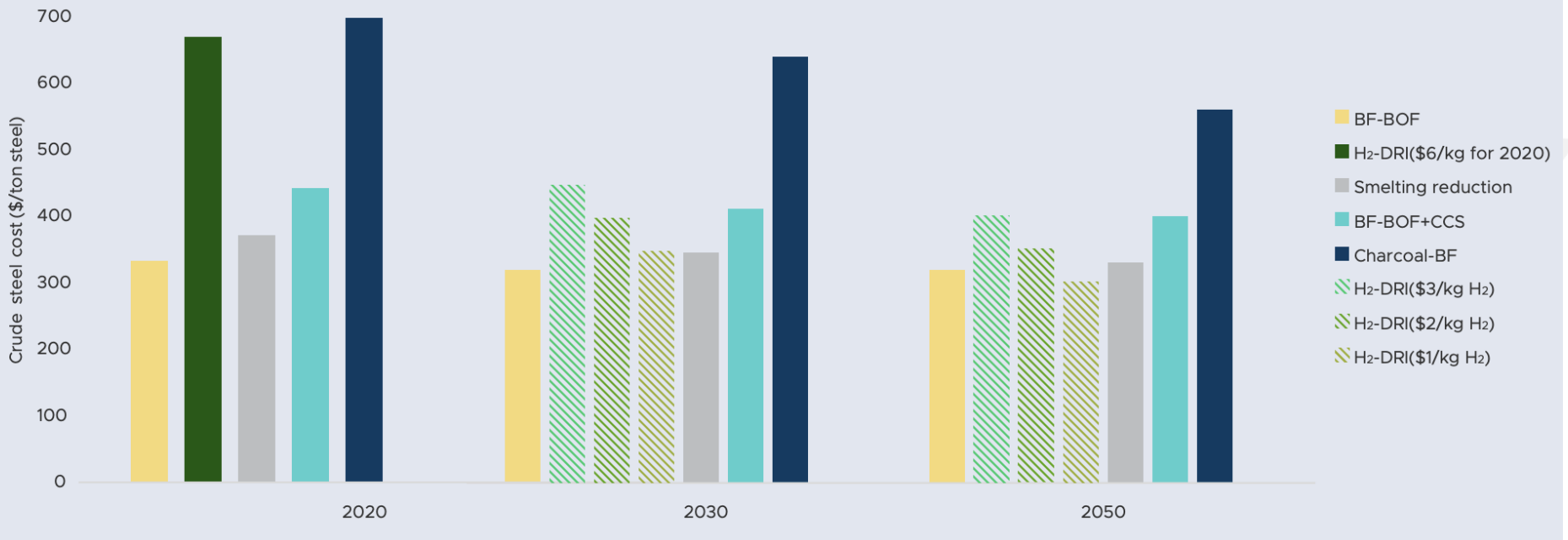
- Grid power backup is the most important strategy, reducing the cost by 12.5% compared to Ex_strat scenario.
- Combining all strategies could reduce the cost to $ 483/t (16.5%), cost parity with BF-BOF!!!



Note: Crude steel cost of BF-BOF is $ 460~500/t based on our calculation

Are we overoptimistic?

- RMI: Even if with $2/kg hydrogen, H2-DRI is still less competitive than BF-BOF by 2050
- Reasons: (1) use low-cost coal and iron ore; (2) without considering flexibility and backups, which are features of new system design and operational mode in the steel industry.



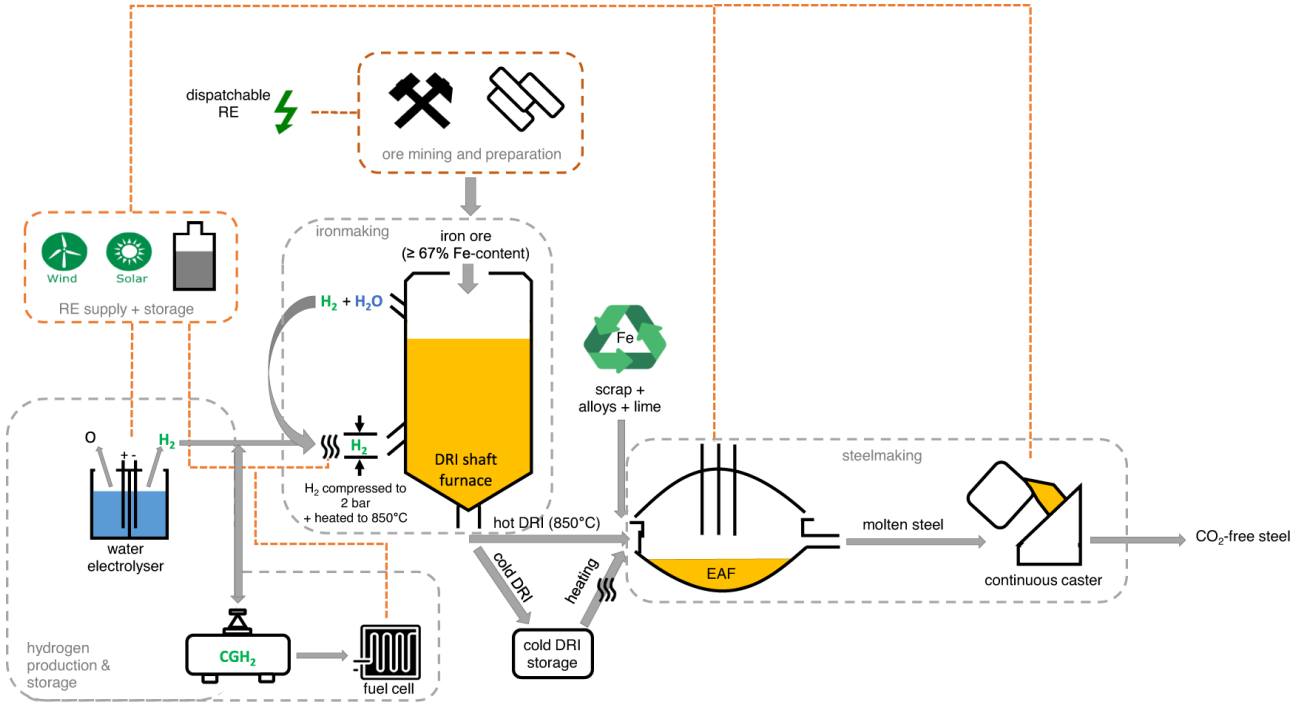
Assumptions:			
Electrolyzer CAPEX	$300/kW	$200/kW	$100/kW
Electricity price	$80/kWh	$50/kWh	$30/kWh
Coking coal price	$200/ton		
Iron ore price	$90/ton		

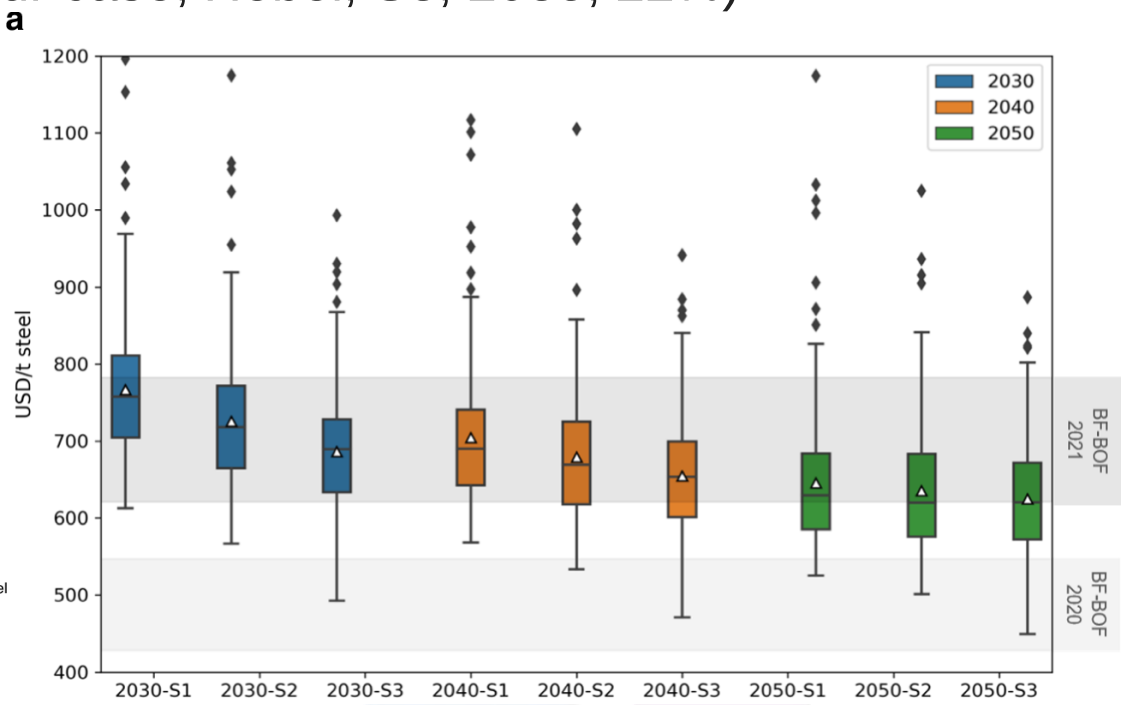
* 2019 feedstock prices are used. The coal and iron ore prices have risen since 2020.

Source: https://rmi.org/insight/pursuing-zero-carbon-steel-in-china/

Are we overoptimistic?

- Paper in *Nature Communication*: with flexibility, could achieve cost parity with high-cost BF-BOF. A significant gap remains compared to the current BF-BOF cost.
- Reason: without considering all strategies (DRI, scrap hourly ratio, steel demand, gas backup, grid backup). $ 619/t (CN, S1, 2050) → $ 483/t (Our case, Hebei, S3, 2035, 22%)





Steel production costs, at site (inc. ore) (case S1 = 0% scrap, S2 = 25% scrap, S3 = 50% scrap)

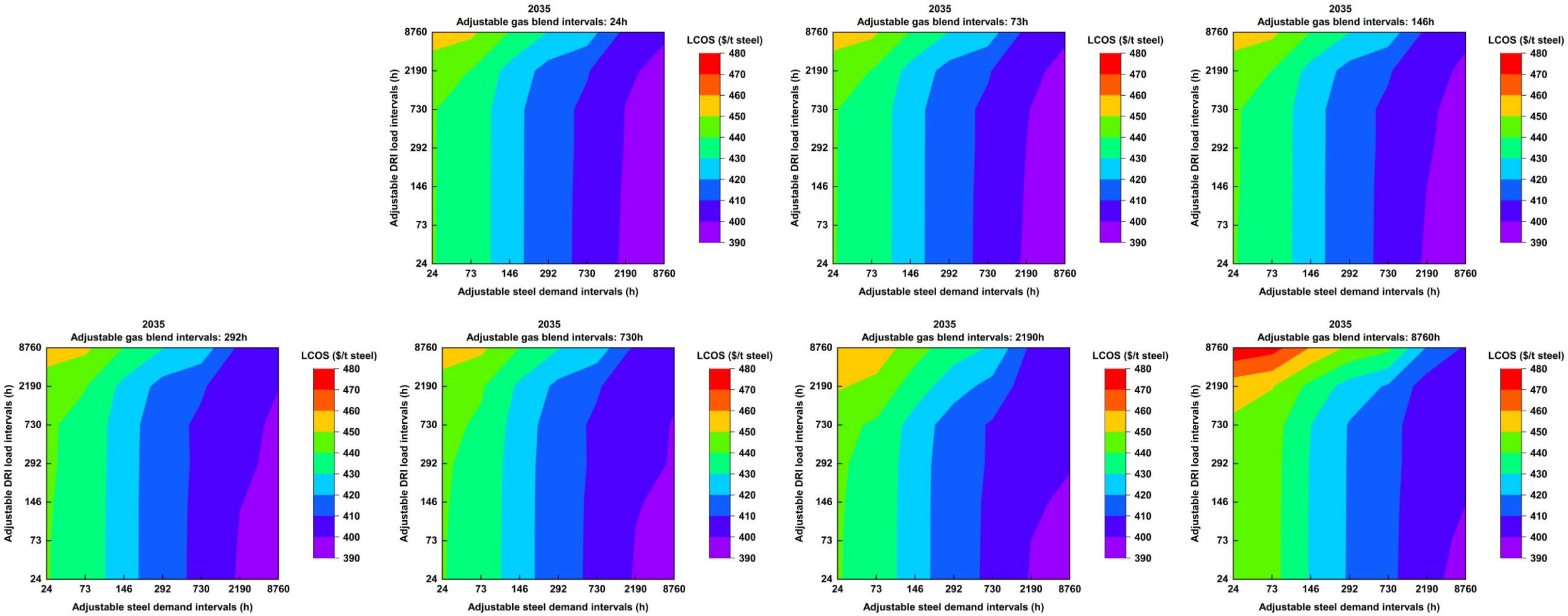
Source: Devlin, A., Kossen, J., Goldie-Jones, H. *et al.* Global green hydrogen-based steel opportunities surrounding high quality renewable energy and iron ore deposits. *Nat Commun* **14**, 2578 (2023). https://doi.org/10.1038/s41467-023-38123-2

Should we maximize flexibility?

- H2 & gas blend ratio changing rate (similar to ramp): higher flexibility (like changing by hour) vs operation complexity and maintenance cost
- DRI load ramp: higher flexibility (like changing by hour) vs operation complexity and maintenance cost
- Grid power: higher flexibility (like changing by hour) vs the tolerance of power grid
- Steel demand: how to define the demand period, which means during the period the total supply must meet the total average demand (hourly, weekly, monthly or annually?)
- **Definitions & assumptions for adjustable intervals:**
  - For gas blend, DRI load, and grid power, values should remain fixed within each interval (intra-interval) but may vary between intervals (inter-interval)
  - For steel demand, values should remain fixed between intervals (inter-interval) but may vary within each interval (intra-interval). The physical meaning is to fulfill the demand specified in the contract over a defined period

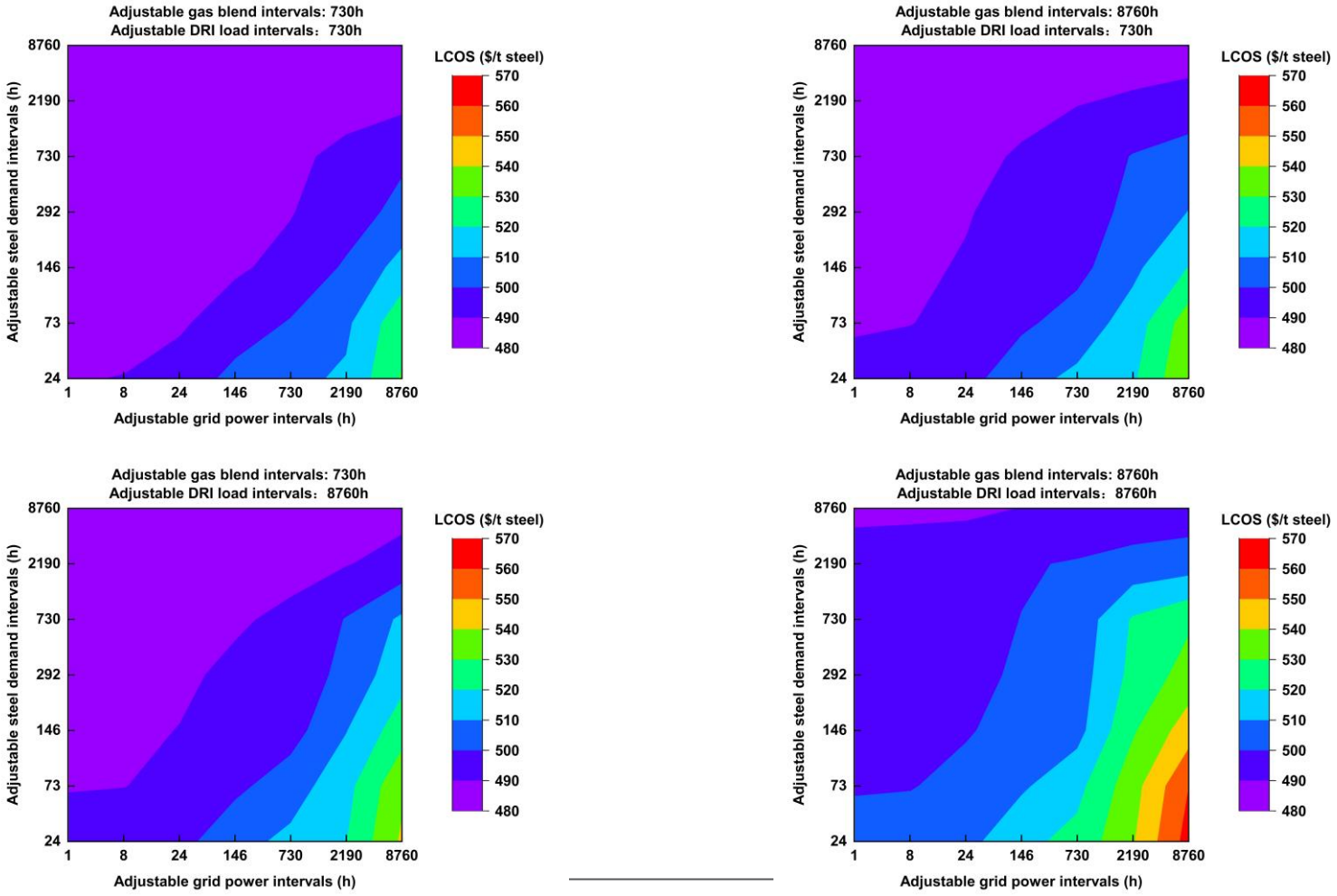
Should we maximize flexibility?

- Steel demand flexibility has greater impact on LCOS than DRI flexibility (blend and load)
- DRI load adjustable interval: monthly is enough; gas blend: monthly or seasonally



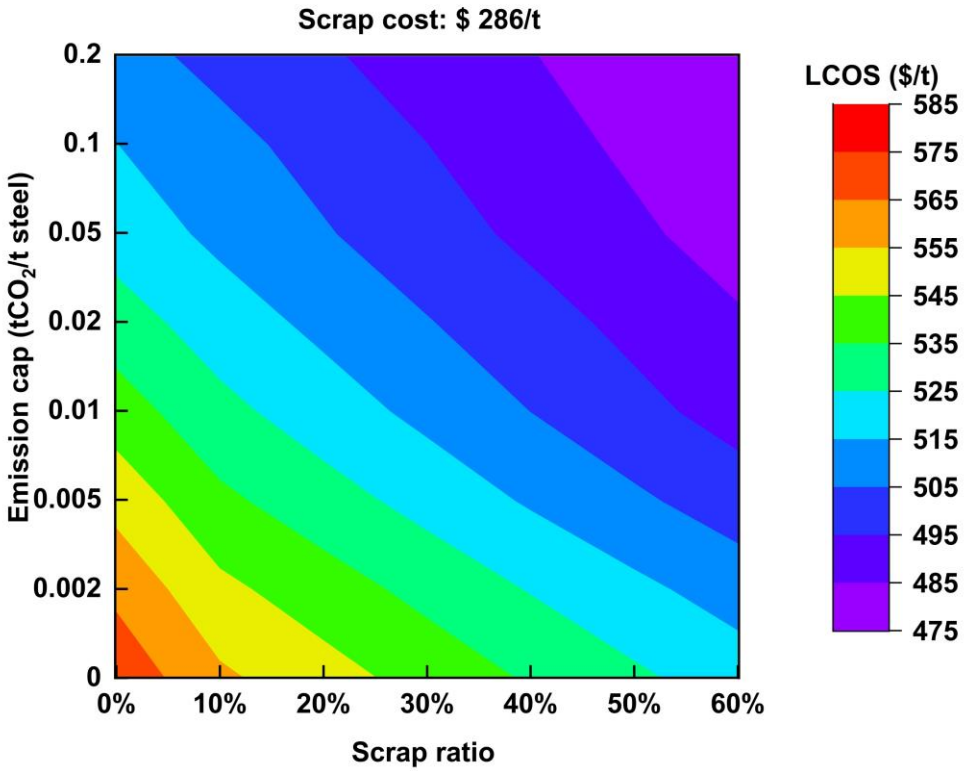
Should we maximize flexibility?

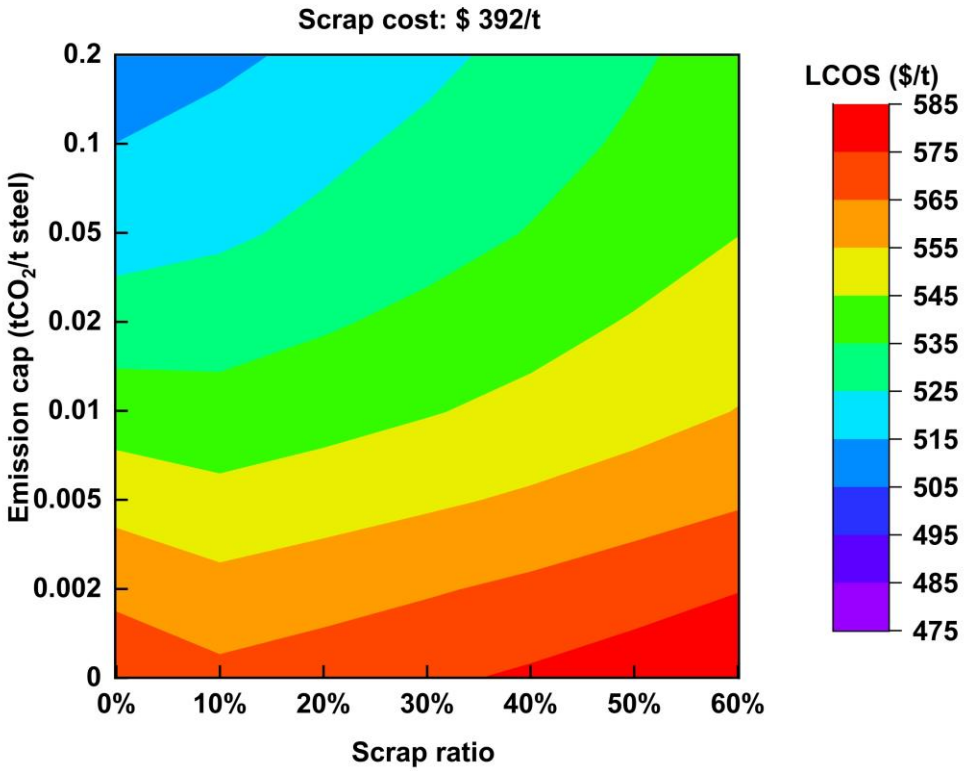
- Grid power intervals: Using 8-hour or 24-hour would not significantly increase costs compared to hourly intervals, and the impact can be further minimized with increased flexibility in steel demand



Balance decarbonization target and cost

- Using China's current scrap price cannot help reduce the cost of the flexible steelmaking system, which shows the importance of developing the scrap recycling system.
- The increase of the LCOS could be observed with tighter decarbonization targets, especially when the scrap is insufficient under low scrap cost.





Note: Crude steel cost of BF-BOF is $ 460~500/t based on our calculation

Balance decarbonization target and cost

Source: https://en.macromicro.me/series/875/scrap-steel-spot

Thank You For Your Attention!

Yuezhang HE
PhD candidate
Department of Energy and Power Engineering
Tsinghua University
Email: heyz21@mails.tsinghua.edu.cn